

Chapter 5: The Real Numbers

Analysis I, by Terrence Tao

A: Cauchy Sequences

Definition 1 - (Eventual) ε -steadiness

Let $\varepsilon > 0$. A sequence $(a_n)_{n=0}^\infty$ is ε -steady if and only if for all natural numbers j, k , $d(a_j, a_k) \leq \varepsilon$. A sequence $(a_n)_{n=0}^\infty$ is eventually ε -steady if and only if the subsequence $(a_n)_{n=N}^\infty$ is ε -steady for some $N \in \mathbb{N}$.

Problem 1 - Examples and Non-examples of (eventual) ε -steadiness

We show examples and non-examples of (eventual) ε -steady sequences.

Proposition 1.1. *The sequence $(a_n) = 10^{-n} = 0.1, 0.01, 0.0001, \dots$ is 0.1-steady but not 0.01-steady.*

Proof. For the sequence $(a_n) = 10^{-n} = 0.1, 0.01, 0.0001, \dots$, observe that for any positive natural numbers j, k we have that $0 \leq 10^{-j}, 10^{-k} \leq 0.1$. Notice that $|10^{-j} - 10^{-k}| \leq \max\{10^{-j}, 10^{-k}\} \leq 0.1$. Hence, the sequence is 0.1-steady.

However, the sequence is not 0.01-steady because $|a_1 - a_2| = |0.1 - 0.01| = 0.09 \geq 0.01$. $\square$

Proposition 1.2. *The sequence $(a_n) = 2^n = 1, 2, 4, 8, \dots$ is not ε -steady for any $\varepsilon > 0$.*

Proof. Suppose for the sake of a contradiction that the sequence $(a_n) = 2^n = 1, 2, 4, 8, \dots$ is ε -steady for some $\varepsilon_0 > 0$. Then, it would imply that for all positive numbers j, k , we have $|2^j - 2^k| \leq \varepsilon_0$. Pick $k = 0$, so $|2^j - 1| \leq \varepsilon_0$ implies $2^j \leq 1 + \varepsilon_0$. Then, by Proposition 4.4.1, we can find a natural number N such that $1 + \varepsilon_0 < N$. Hence, we have that $2^j \leq 1 + \varepsilon_0 < N$, so $2^j < N$. If we choose $j := N$, we have that $2^N < N$, which contradicts the fact that $2^N \geq N$ for all natural numbers N . $\square$

Proposition 1.3. *The sequence $10, 0, 0, 0, \dots$ is not ε -steady for any $\varepsilon < 10$, but is eventually ε -steady for $\varepsilon > 0$.*

Proof. The sequence $10, 0, 0, 0, \dots$ is not ε -steady for any $\varepsilon < 10$ because we have that $|10 - 0| = 10 \not\leq \varepsilon$. However, the sequence is eventually ε -steady if we choose $N = 1$. This is because for all $j, k \geq 1$, we have $|a_j - a_k| = 0 < \varepsilon$. $\square$

Definition 2 - Cauchy Sequences

A sequence $(a_n)_{n=0}^\infty$ is a Cauchy sequence if and only if for all $\varepsilon > 0$, the sequence is eventually ε -steady. In other words, $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $j, k \in \mathbb{N} \implies d(a_j, a_k) \leq \varepsilon$.

Definition 3 - Boundedness

Let $M \geq 0$ be rational. A finite sequence $a_1, a_2, \dots, a_n$ is bounded by M if and only if $|a_i| \leq M$ for all $1 \leq i \leq n$. An infinite sequence is bounded by M if and only if $|a_i| \leq M$ for all $i \geq 0$. Overall, a sequence is said to be bounded if and only if it is bounded by M for some rational $M \geq 0$.

Proposition 1.4. *The sequence $(a_n)_{n=0}^\infty = (-1)^n \times n = 1, -2, 3, -4, \dots$ is unbounded.*

Proof. Suppose the sequence $(a_n)_{n=0}^\infty = (-1)^n \times n = 1, -2, 3, -4, \dots$ is bounded, so there exists a rational number $M \geq 0$ such that $|a_n| = n \leq M$ for all $n \geq 1$. By Proposition 4.4.1, there exists a natural number N such that $N > M$. But by our supposition, we also have that $|a_N| = N \leq M$, a contradiction! $\square$

Problem 2 - Cauchy sequences are bounded

Every Cauchy sequence $(a_n)_{n=0}^\infty$ is bounded.

Proof. Suppose the sequence $(a_n)_{n=0}^\infty$ is a Cauchy sequence, so we choose $\varepsilon = 1$. Then, there exists a natural number N such that for all $j, k \geq N$, we have $d(a_j, a_k) = |a_j - a_k| \leq 1$. Then, by the reverse triangle inequality, $||a_j| - |a_k|| \leq |a_j - a_k| \leq 1$. By simplifying, we get that $|a_j| \leq 1 + |a_k|$. Pick $k := N$ so that for any $j \geq N$, we have $|a_j| \leq 1 + |a_N|$. Therefore, for the subsequence of $(a_n)_{n=N}^\infty$, it is bounded by $1 + |a_N|$. Then, for the finite subsequence $(a_n)_{n=0}^N$, there exists M such that M bounds the finite subsequence. Therefore, the whole sequence is bounded by $\max\{1 + |a_N|, M\}$. $\square$

B: Equivalent Cauchy Sequences

Definition 4 - Equivalent Sequences

Two sequences $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are equivalent if and only if for every $\varepsilon > 0$, the sequences are eventually ε -close. In other words, $\exists N \in \mathbb{N}$ such that $\forall n \geq N$, $|a_n - b_n| < \varepsilon$.

Problem 1 - Equivalence of Cauchy Sequences

If $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are equivalent sequences of rationals, then $(a_n)_{n=0}^\infty$ is a Cauchy sequence if and only if $(b_n)_{n=0}^\infty$ is a Cauchy sequence.

Proof. Suppose $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are equivalent sequences of rationals, and suppose also $(a_n)_{n=0}^\infty$ is a Cauchy Sequence. We now show that $(b_n)_{n=0}^\infty$ is also Cauchy. Let $\varepsilon > 0$ be given.

By the equivalence of the two sequences, there exists a natural number N such that if $n \geq N_1$, then $d(b_n, a_n) \leq \frac{\varepsilon}{4}$. Moreover, we also have by $(a_n)_{n=0}^\infty$ being Cauchy that there exists a natural number M such that if $j, k \geq M$, then $d(a_j, a_k) \leq \frac{\varepsilon}{2}$. We choose $L = \max\{N, M\}$ so that we have the useful properties from equivalence and Cauchy sequences. Then, by one of the properties of the distance function (specifically of “If $d(x, y) \leq \varepsilon$ and $d(z, w) \leq \delta$, then $d(x \pm z, y \pm w) \leq \varepsilon + \delta$ ”), we have that

$$\begin{aligned}
d(b_j, a_j) \leq \frac{\varepsilon}{4} \text{ and } d(b_k, a_k) \leq \frac{\varepsilon}{4} &\implies d(b_j - b_k, a_j - a_k) \leq \frac{\varepsilon}{2} && \text{by properties of the distance function} \\
&\implies |(b_j - b_k) - (a_j - a_k)| \leq \frac{\varepsilon}{2} && \text{by definition of distance} \\
&\implies ||b_j - b_k| - |a_j - a_k|| \leq \frac{\varepsilon}{2} && \text{by the reverse triangle inequality} \\
&\implies |b_j - b_k| - |a_j - a_k| \leq \frac{\varepsilon}{2} && \text{because we only care about being } \leq \frac{\varepsilon}{2} \\
&\implies |b_j - b_k| - \frac{\varepsilon}{2} \leq \frac{\varepsilon}{2} && \text{by } (a_n)_{n=0}^\infty \text{ being a Cauchy sequence} \\
&\implies d(b_j, b_k) \leq \varepsilon && \text{by rearrangement.}
\end{aligned}$$

For the converse, just switch the two sequences. $\square$

Problem 2 - Boundedness of Equivalent Cauchy Sequences

Let $\varepsilon > 0$. Show that if $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are eventually ε -close, then $(a_n)_{n=0}^\infty$ is bounded if and only if $(b_n)_{n=0}^\infty$ is bounded.

Proof. Let $\varepsilon > 0$ be given. Suppose the sequences $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are eventually ε -close, so they are equivalent sequences. Suppose also that $(a_n)_{n=0}^\infty$ is bounded, so there exists a non-negative rational M such that $|a_i| \leq M$ for all $i \geq 0$. We wish to show that $(b_n)_{n=0}^\infty$ is also bounded.

Because the two sequences are eventually ε -close, without loss of generality, we arbitrarily picked $\varepsilon := 1$. We then choose N such that for all $n \geq N$, we have that $|b_n - a_n| \leq 1$. Then, by the reverse triangle inequality, we have that $||b_n| - |a_n|| \leq |b_n - a_n| \leq 1$. By transitivity, we get $||b_n| - |a_n|| \leq 1$. By the properties of the absolute value, we have that $-1 \leq |b_n| - |a_n| \leq 1$. Focusing on the middle and right hand side of the inequalities, we get $|b_n| - |a_n| \leq 1$. Adding $|a_n|$ to both sides gives $|b_n| \leq 1 + |a_n|$. Because of $(a_n)_{n=0}^\infty$ being bounded, we have that $|a_n| \leq M$. By substitution, we get $|b_n| \leq 1 + M$. Therefore, for the tail of the infinite subsequence $(b_n)_{n=N}^\infty$, it is bounded by $M + 1$. For the head of the finite subsequence, it is bounded by some $L := \max\{|b_0|, |b_1|, \dots, |b_{N-1}|\}$. Hence, the overall subsequence $(b_n)_{n=0}^\infty$ is bounded by $\max\{1 + M, L\}$.

For the converse, just switch the sequences being treated. The logic follows because the distance function is symmetric. $\square$

C: The Construction of Real Numbers

Definition 5 - Real Numbers ($\mathbb{R}$)

A real number is defined to be an object of the form $\text{LIM}_{n \rightarrow \infty} a_n$, where $(a_n)_{n=0}^{\infty}$ is a Cauchy sequence of rational numbers. Two real numbers $\text{LIM}_{n \rightarrow \infty} a_n$ and $\text{LIM}_{n \rightarrow \infty} b_n$ are said to be equal if and only if $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ are equivalent Cauchy sequences.

For all our exercises in this section, let $x = \text{LIM}_{n \rightarrow \infty} a_n$, $y = \text{LIM}_{n \rightarrow \infty} b_n$, and $z = \text{LIM}_{n \rightarrow \infty} c_n$ be real numbers.

Problem 1 - Formal limits are well-defined

Our definition of equality follows the three axioms of reflexivity, symmetry, and transitivity.

Proof. To show reflexivity, consider the real number x . Let $\varepsilon > 0$ be given. Choose $N = 0$. Then, for all $n \geq N$, we have $|a_n - a_n| = 0 \leq \varepsilon$. Hence, $x = x$.

To show symmetry, suppose $x = y$, so for all $\varepsilon > 0$, there exists a natural number N such that for all $n \geq N$, we have $d(a_n, b_n) \leq \varepsilon$. By the symmetry of distance function we have $d(b_n, a_n) \leq \varepsilon$. Hence, $y = x$.

To show reflexivity, let $\varepsilon > 0$ be given. Suppose $x = y$ and $y = z$, so there exist natural numbers N_1 and N_2 such that for all $n \geq N_1$, we have $d(a_n, b_n) \leq \frac{\varepsilon}{2}$ and for all $n \geq N_2$, we have $d(b_n, c_n) \leq \frac{\varepsilon}{2}$. Then, by the triangle inequality, we get for all $n \geq \max\{N_1, N_2\}$ that

$$d(a_n, c_n) \leq d(a_n, b_n) + d(b_n, c_n) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \leq \varepsilon.$$

Hence, $x = z$. □

Definition 6 - Multiplication for the Real Numbers

The product xy is defined to be $xy := \text{LIM}_{n \rightarrow \infty} a_n b_n$.

Problem 2 - Multiplication is well-defined.

Let $x' = \text{LIM}_{n \rightarrow \infty} a'_n$. The product xy is a real number. Also, if $x = x'$, then $xy = x'y$.

Proof. Let $\varepsilon > 0$ be given.

First, we show that xy is a real number. Let $\delta_1, \delta_2 > 0$. Since $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ are Cauchy, there exist natural numbers N_1, N_2 such that for all $j, k \geq N_1$, we have $d(a_j, a_k) \leq \delta_1$ AND for all $j, k \geq N_2$, we also have that $d(b_j, b_k) \leq \delta_2$. If $j, k \geq \max\{N_1, N_2\}$, then by Proposition 4.3.7h, we have $d(a_j b_j, a_k b_k) \leq |a_j| \delta_2 + |b_j| \delta_1 + \delta_1 \delta_2$.

Now, our goal is to bound the right hand side of the inequality to be less than or equal to ε . By the property of Cauchy sequences, they are bounded, so there exist positive rationals M_1 and M_2 such that $|a_i| \leq M_1$ and $|b_i| \leq M_2$ for any natural number i . Choose $M = \max\{M_1, M_2\}$ so that $|a_i| \leq M$ and $|b_i| \leq M$ for any natural number i .

Now, set $\delta_1 = \delta_2 = \delta$ where $\delta := \min\{1, \frac{\varepsilon}{3M}\}$. Then, we can find a natural number N such that for all $j, k \geq N$, we have $d(a_j, a_k) \leq \delta$ and $d(b_j, b_k) \leq \delta$. (We do not delineate between N_1 and N_2 for the two sequences. Instead, we just choose the maximum.) Since $\delta \leq \frac{\varepsilon}{3M}$, we get by rearrangement that $M\delta \leq \frac{\varepsilon}{3}$. Moreover, notice that $\delta \leq 1$. Hence, $\delta^2 \leq \delta \leq \frac{\varepsilon}{3M} \leq \frac{\varepsilon}{3}$. Overall, we have by Proposition 4.3.7h:

$$\begin{aligned} d(a_j b_j, a_k b_k) &\leq |a_j| \delta + |b_j| \delta + \delta \delta \\ &\leq M\delta + M\delta + \delta^2 \\ &\leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &\leq \varepsilon \end{aligned}$$

Second, we show that multiplication is well-defined. Let $\varepsilon > 0$ be given. Suppose $x = x'$. By the property of Cauchy sequences, $(b_n)_{n=0}^{\infty}$ is bounded by some positive rational M (i.e., $|b_i| \leq M$ for any

natural number i). By the equivalence of x and x' , we have for $\delta := \frac{\varepsilon}{M}$ that there exists a positive natural number N such that for all $n \geq N$, we have $d(a_n, a'_n) \leq \delta$. Then, for all $n \geq N$, we have:

$$\begin{aligned} d(a_n b_n, a'_n b_n) &= |a_n b_n - a'_n b_n| \\ &= |b_n(a_n - a'_n)| \\ &= |b_n| |a_n - a'_n| \\ &\leq M |a_n - a'_n| \\ &\leq M \left(\frac{\varepsilon}{M} \right) \\ &= \varepsilon \end{aligned}$$

Overall, we have that $d(a_n b_n, a'_n b_n) \leq \varepsilon$ for all $n \geq N$. Therefore, $xy = x'y$. $\square$

Definition 7 - Negation and Subtraction

We define the negation $-x$ as $-x := (-1) \times x$. We define the difference of x and y as $x - y := x + (-y) = x + (-1) \times y$.

Problem 3 - Implications of Negation and Subtraction

Show $-\text{LIM}_{n \rightarrow \infty} a_n = \text{LIM}_{n \rightarrow \infty} -a_n$. Show also $\text{LIM}_{n \rightarrow \infty} a_n - \text{LIM}_{n \rightarrow \infty} b_n = \text{LIM}_{n \rightarrow \infty} (a_n - b_n)$.

Proof. For the first equation about negation, notice:

$$\begin{aligned} -(\text{LIM}_{n \rightarrow \infty} a_n) &= (-1) \times \text{LIM}_{n \rightarrow \infty} a_n && \text{by definition of negation} \\ &= \text{LIM}_{n \rightarrow \infty} -1 \times \text{LIM}_{n \rightarrow \infty} a_n && \text{by embedding the rationals to the reals} \\ &= \text{LIM}_{n \rightarrow \infty} ((-1) \times a_n) && \text{by definition of multiplication over the reals} \\ &= \text{LIM}_{n \rightarrow \infty} -a_n && \text{by the property that } \forall x \in \mathbb{Q}, (-1) \times x = -x \end{aligned}$$

For the second equation about subtraction, notice:

$$\begin{aligned} \text{LIM}_{n \rightarrow \infty} a_n - \text{LIM}_{n \rightarrow \infty} b_n &= \text{LIM}_{n \rightarrow \infty} a_n + \text{LIM}_{n \rightarrow \infty} -b_n && \text{by } -\text{LIM}_{n \rightarrow \infty} a_n = \text{LIM}_{n \rightarrow \infty} -a_n \\ &= \text{LIM}_{n \rightarrow \infty} a_n - b_n && \text{by definition of addition on the reals} \end{aligned}$$

$\square$

Problem 4 - Embedding the Rational Numbers to the Reals

Let a, b be rational numbers. Show that $a = b$ if and only if $\text{LIM}_{n \rightarrow \infty} a = \text{LIM}_{n \rightarrow \infty} b$ (i.e., the Cauchy sequences $(a_n)_{n=0}^\infty = a, a, a, \dots$ and $(b_n)_{n=0}^\infty = b, b, b, \dots$ are equivalent if and only if $a = b$.) This allows us to embed the rational numbers inside the real numbers in a well-defined manner.

Proof. Suppose $a = b$. Let $\varepsilon > 0$ be given. Notice that the sequences $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are equivalent because we can choose $N = 0$ so that for all $n \geq N$, we have $|a_n - b_n| = |a - b| = |a - a| = 0 \leq \varepsilon$.

Conversely, suppose instead that $\text{LIM}_{n \rightarrow \infty} a_n = \text{LIM}_{n \rightarrow \infty} b_n$. Recall Proposition 4.3.7a, which states that $\forall x, y \in \mathbb{Q}, x = y \iff \forall \varepsilon > 0, d(x, y) \leq \varepsilon$. Because for every $\varepsilon > 0$, there exists a natural number N such that for all $n \geq N$, we have $d(a_n, b_n) = d(a, b) \leq \varepsilon$, we have that $a = b$. $\square$

Problem 5 - Boundedness of Equivalent Cauchy Sequences (But Again???)

Let $(a_n)_{n=0}^\infty$ be a sequence of rational numbers, which is bounded. Let $(b_n)_{n=0}^\infty$ be another sequence of rational numbers, which is also equivalent to $(a_n)_{n=0}^\infty$. Show that $(b_n)_{n=0}^\infty$ is also bounded.

Proof. This problem is just a restatement from what has been proven beforehand. Let $\varepsilon > 0$. Suppose $(a_n)_{n=0}^\infty$ is equivalent to $(b_n)_{n=0}^\infty$, so the sequences are eventually ε -close. We have already shown beforehand that if $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are eventually ε -close, then $(a_n)_{n=0}^\infty$ is bounded if and only if $(b_n)_{n=0}^\infty$ is bounded. Then, because $(a_n)_{n=0}^\infty$ is bounded, then $(b_n)_{n=0}^\infty$ is also bounded. $\square$

Problem 6 - $\text{LIM}_{n \rightarrow \infty} \frac{1}{n} = 0$.

Show that $\text{LIM}_{n \rightarrow \infty} \frac{1}{n} = 0$.

Proof. Since we can embed the rational numbers to the reals, we have that $0 = \text{LIM}_{n \rightarrow \infty} 0$. It would be enough to show that $\text{LIM}_{n \rightarrow \infty} \frac{1}{n} = \text{LIM}_{n \rightarrow \infty} 0$. Now, let $\varepsilon > 0$ be given. Choose a natural number N such that $N > \frac{1}{\varepsilon}$ (i.e., $\frac{1}{N} < \varepsilon$). We know this N exists because of Proposition 4.4.1, which states that $\forall x \in \mathbb{Q}, \exists N \in \mathbb{N}$ such that $N > x$. Notice that for all $n \geq N$, we have $d(\frac{1}{n}, 0) = |\frac{1}{n} - 0| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. Therefore, $\text{LIM}_{n \rightarrow \infty} \frac{1}{n} = 0$ because $d(\frac{1}{n}, 0) \leq \varepsilon$ for all $n \geq N$. $\square$

Definition 8 - Reciprocation of real numbers

A sequence $(a_n)_{n=0}^{\infty}$ of rational numbers is said to be bounded away from zero if and only if there exists a rational number $c > 0$ such that $|a_n| \geq c$ for all $n \geq 1$.

Let x be a non-zero real number. Let $(a_n)_{n=0}^{\infty}$ be a Cauchy sequence bounded away from zero such that $x = \text{LIM}_{n \rightarrow \infty} a_n$ (such a sequence exists by Lemma 5.3.14). Then, we define the reciprocal x^{-1} by the formula $x^{-1} := \text{LIM}_{n \rightarrow \infty} a_n^{-1}$. (From Lemma 5.3.15, we know that x^{-1} is a real number.)

Finally, we define division x/y of two real numbers x, y , provided y is non-zero by the formula:

$$x/y := x \times y^{-1}.$$

Problem 7 - Multiplicative Inverse

Show that $xx^{-1} = x^{-1}x = 1$.

Proof. Notice:

$$\begin{aligned} xx^{-1} &= \text{LIM}_{n \rightarrow \infty} a_n \times \text{LIM}_{n \rightarrow \infty} a_n^{-1} && \text{by definition of } x \text{ and } x^{-1} \\ &= \text{LIM}_{n \rightarrow \infty} a_n \times a_n^{-1} && \text{by definition of multiplication over the reals} \\ &= \text{LIM}_{n \rightarrow \infty} 1 && \text{by the multiplicative inverse of the rationals} \\ &= 1 && \text{by the embedding of } \mathbb{Q} \text{ to } \mathbb{R} \end{aligned}$$

Because multiplication is commutative, we also have $x^{-1}x = 1$. $\square$